

Linux Basics for DevOps

Growtech Digital Solutions – Training Resource (September 2025)

Why Linux for DevOps?

Linux is the foundation of modern DevOps practices. It powers most servers, cloud infrastructure, and container platforms. Mastering Linux is a must for any DevOps engineer.

- Most CI/CD and automation tools run on Linux.
- Containers (Docker, Kubernetes) rely heavily on Linux internals.
- Cloud providers (AWS, Azure, GCP) use Linux-based virtual machines.

Key Linux Skills for DevOps

File System & Navigation: ``ls``, ``cd``, ``pwd``, ``mkdir``, ``rm`` – Understand absolute vs relative paths.

User & Permissions Management: ``chmod``, ``chown``, ``sudo`` – Set permissions to control access.

Process Management: ``ps``, ``top``, ``kill`` – Monitor and stop processes.

Networking Basics: ``ping``, ``netstat``, ``curl``, ``dig`` – Check connectivity and troubleshoot issues.

Shell Scripting: Automate tasks using Bash scripts. Variables, loops, and conditional statements.

Practical Tips

- Practice on a Linux VM or cloud instance.
- Use automation to solve small problems.
- Learn how to install packages (apt, yum, dnf).
- Document your commands and create scripts.

Conclusion

Linux is the first step in a DevOps journey. Once you master the basics, you can progress to tools like Docker, Kubernetes, and Terraform with confidence.

For training & guidance: growtechdigitalsolutions@gmail.com